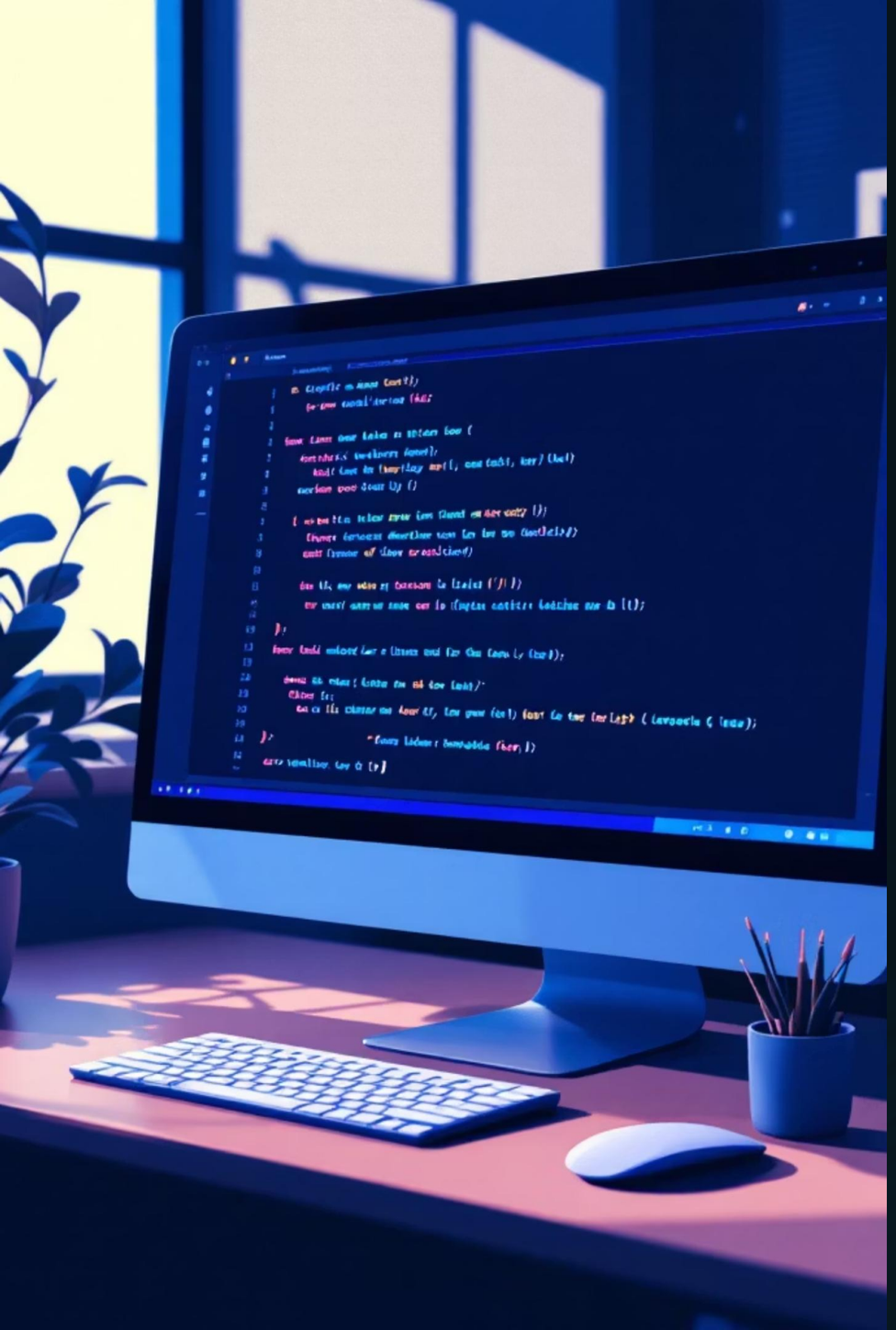
Sales Performance Analysis

Comprehensive retail data analysis uncovering revenue trends, product performance, and profit optimization opportunities

Dataset Overview

200	14	0
Transactions	Features	Missing Values
Rows of sales data	Columns analyzed	Clean dataset

Complete transaction records with sales details, product insights, regional distribution, and profitability metrics



Python Data Preparation

Data Loading

Imported using pandas with initial exploration via `df.info()` and `describe()`

Feature Engineering

Created Profit and Profit (%) metrics for enhanced analysis

Standardization

Cleaned column names, exported to SQL database

Total Revenue & Profit

Total Product Cost

INR

Total Sales

INR

Total Profit

INR

Strong profitability with 29.58% profit margin across all transactions

Category Performance

Electronics

₹21.39L profit • 35.57% margin

Furniture

₹11.09L profit • 30.69% margin

Accessories

₹1.59L profit • 66.18% margin

Electronics drives revenue while Accessories shows highest margin percentage

Regional Sales Performance

29.6%

North

₹2.53M sales

25.1%

East

₹2.69M sales

25.1%

South

₹2.86M sales

30.1%

West

₹3.44M sales · Highest profit

Top Products & Customers

Best Sellers

- Tablet: ₹12.75L
- Table: ₹9.15L
- Headphones: ₹2.90L
- Chair: ₹2.22L
- Mouse: ₹70K

Top Customers

- Rahul: ₹20.12L (28 orders)
- Suresh: ₹14.17L (19 orders)
- Meena: ₹12.94L (21 orders)
- Kiran: ₹12.04L (24 orders)

Monthly Trends

1. **January**
   ₹2.12M • 28.00%
2. **February**
   ₹2.37M • 28.18%
3. **March**
   ₹1.88M • 27.53%
4. **April**
   ₹1.62M • 29.62%
5. **May**
   ₹2.02M • 30.16%
6. **June**
   ₹1.51M • 31.94%

February shows highest sales volume; June achieves peak profitability at 31.94%

Key Insights

No Loss Products

All items generate positive profit

High-Margin Accessories

66.18% margin despite lower revenue

West Region Leader

Highest sales and profitability

Revenue-Profit Disconnect

High sales don't always equal high profit

Strategic Recommendations

01

Prioritize Electronics

Focus marketing on high-revenue category

02

Boost Regional Sales

Target North and East regions

03

Optimize Product Mix

Emphasize top performers, review low performers

04

Seasonal Campaigns

Leverage February and May peaks

05

Improve Margins

Optimize pricing on mid-performing products

06

Enhance Bundling

Encourage cross-selling with Accessories